

Assignment Cover Sheet	
Candidate Number	315663
Module Code	BEMM462_RWA_1_202526
Module Name	Operations Analytics
Assignment Title	Individual Report.

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University
of Exeter

BEMM462 Individual Report — Operations Analytics

Project 1: Data Envelopment Analysis.

1. Question 1

Resource used by each Decision-Making Unit (DMU) are considered input values whereas the desired outcome of a given DMU is called output values in Data Envelopment Analysis (DEA). Relevant measures according to the eight engineering

projects which are to be considered are as follows:

Person-days, CPU time (hours). These estimate the effort in the labour and computation per project. The more person-days, the more human labour, the more CPU time, the more computer resources would be used extensively. They are all expenses to the firm that should be reduced (Cooper et al., 2007). Profit (million £). This measure accounts the generated value of every project thereby a greater profit is desired. This classification is aligned with the concept of efficiency of DEA, i.e. determination of the ratio of (weighted) outputs to (weighted) inputs. Higher efficiency is indicated by a higher rate of return on a unit of resource. From a Manager's perspective, consulting companies are interested in getting maximum when it comes to the amount of labour and computing resources allocated to a project.

2. Question 2

CCR (Charnes, Cooper, and Rhodes) model is used. The present model will be implemented with output-based strategies taking into consideration the CRS. This model is applicable to all DMUs with same conditions. As an example, a consulting firm can have a number of projects on the same (very) similar operational basis (Charnes et al., 1978) project pipeline.

Problem Formulation: Maximize θ = Virtual Output / Virtual Input.

Constraints:

- Virtual Input of the target DMU = 1 (Charnes-Cooper normalization).
- Virtual Input $\geq$ Virtual Output for every DMU.
- All weights > 0 .

Using the Excel Solver, the LP approach was used to address this challenge. As DEA needs a solution of an LP per DMU, the project target chooser (in B15) varied 1 through 8. Due to the fact that every change implied that the solver was reused, the 8 projects had different reports about answers. The summary of the overall project results is presented below.

B17 {=SUMPRODUCT(--(\$F\$2:\$F\$9=\$B\$15), \$I\$2:\$I\$9)}										
	A	B	C	D	E	F	G	H	I	J
1	DATA SET	Person-days	CPU Time	Profit		PROJECT	VIRTUAL	SIGN	VIRTUAL	
2	Project 1	550	200	2.1		1	2.75	>=	1.05	
3	Project 2	400	150	0.5		2	2	>=	0.25	
4	Project 3	300	400	3		3	1.5	>=	1.5	
5	Project 4	350	450	2		4	1.75	>=	1	
6	Project 5	450	300	1		5	2.25	>=	0.5	
7	Project 6	500	150	1.5		6	2.5	>=	0.75	
8	Project 7	350	200	0.6		7	1.75	>=	0.3	
9	Project 8	200	600	1.8		8	1	>=	0.9	
10										
11	DECISION VARS	Wgt Input	Wgt	Wgt		Project	θ	Status		
12	Optimal Weights	0.005		0	0.5	P1	1.0000	Efficient		
13						P2	0.3213	Inefficient		
14	TARGET DMU					P3	1.0000	Efficient		
15	Select Project:	8				P4	0.5893	Inefficient		
16	Target Input:	1	=	1		P5	0.3865	Inefficient		
17	Target Output:	0.9	<=			P6	0.9524	Inefficient		
18	Efficiency %:	90.00%			OBJECTIVE	P7	0.3333	Inefficient		
19						P8	0.9000	Inefficient		
20										
21										
22	NOTES — P1 Q1									
23	#	Cell	Note							
24	1	B12	Optimal weight for person-days input (from last Solver run)							
25	2	G12	Summary: P1, P3 efficient. All others inefficient (basic 2-input/1-output CCR DEA)							
26	3	B15	Change this to project (1-8), then re-run Solver to get that project's θ							
27	4	B16	Looks up Virtual Input for the selected project							
28	5	D16	Constraint: target VI = 1 (Charnes-Cooper normalisation)							
29	6	B18	$\theta = VO/VI$ for target DMU. $\theta \geq 1 \rightarrow$ efficient							
30										

Figure 1: P1 Q1 sheet: DEA data, Virtual Input/Output columns, and summary table

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
 Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Figure 2: Solver settings for the CCR model

The project overall has been summarised to give the following results:

Project	Efficiency (θ)	Status
P1	1.0000	Efficient
P2	0.3213	Inefficient
P3	1.0000	Efficient
P4	0.5893	Inefficient
P5	0.3865	Inefficient
P6	0.9524	Inefficient
P7	0.3333	Inefficient
P8	0.9000	Inefficient

F26

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Microsoft Excel 16.0 Answer Report

2

Worksheet: [Operations Individual Assignment.xlsx]P1 Q1

3

Report Created: 3/14/2026 10:38:54 AM

4

Result: Solver found a solution. All Constraints and optimality conditions are satisfied.

5

Solver Engine

6

Engine: Simplex LP

7

Solution Time: 0.031 Seconds.

8

Iterations: 12 Subproblems: 0

9

Solver Options

10

Max Time Unlimited, Iterations Unlimited, Precision 0.000001

11

Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

12

13

14

Objective Cell (Max)

15

Cell	Name	Original Value	Final Value
\$B\$17	Target Output: Wgt Input	0	1

16

17

18

19

Variable Cells

20

Cell	Name	Original Value	Final Value	Integer
\$B\$12	Optimal Weights Wgt Input	0	0.000714286	Contin
\$C\$12	Optimal Weights Wgt	0	0.003035714	Contin
\$D\$12	Optimal Weights Wgt	0	0.476190476	Contin

21

22

23

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26

Constraints

27

Cell	Name	Cell Value	Formula	Status	Slack
\$B\$16	Target Input: Wgt Input	1	\$B\$16=\$D\$16	Binding	0
\$G\$2	Project 1 VIRTUAL INPUT	1	\$G\$2>=\$I\$2	Binding	0
\$G\$3	Project 2 VIRTUAL INPUT	0.741071429	\$G\$3>=\$I\$3	Not Binding	0.50297619
\$G\$4	Project 3 VIRTUAL INPUT	1.428571429	\$G\$4>=\$I\$4	Binding	0
\$G\$5	Project 4 VIRTUAL INPUT	1.616071429	\$G\$5>=\$I\$5	Not Binding	0.663690476
\$G\$6	Project 5 VIRTUAL INPUT	1.232142857	\$G\$6>=\$I\$6	Not Binding	0.755952381
\$G\$7	Project 6 VIRTUAL INPUT	0.8125	\$G\$7>=\$I\$7	Not Binding	0.098214286
\$G\$8	Project 7 VIRTUAL INPUT	0.857142857	\$G\$8>=\$I\$8	Not Binding	0.571428571
\$G\$9	Project 8 VIRTUAL INPUT	1.964285714	\$G\$9>=\$I\$9	Not Binding	1.107142857

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Figure 3: Answer Report for P1 (Efficient)

A1

Figure 4: Answer Report for P3 (Efficient)

P1 and P3 are an efficient project when $\theta = 1.0$, that is why there is no mixture of the input of any other project that obtain at least an equal output with less input. The project P7 also exhibits the least efficiency of all the projects under analysis ($\theta = 0.33$) that means that the theoretical decrease in the total inputs of P7 in terms of the same output would be 67 percent when it functions along the efficient production frontier. Projects P6 and P8 are almost efficient, therefore, just a slight, out of the production frontier

3. Question 3

The final recommendation is project 3. The score of its efficiency is 1.0 and it lies on the efficient frontier. It also makes the greatest profit (£3.0 million) by the comparatively fewest person-days (300) that any of the projects use. Therefore, this is the most effective utilisation of the company resources. The second recommendation is project 1; this is an

efficient project ($\theta = 1.0$) having a profit (£2.1 million). Among the two, project 3 has the highest profit-resource ratio and must be ranked the first to the best.

4. Question 4

The DEA model was supplemented with additional steps in order to enhance the recommendation. Two further dimensions are brought into play, Budget (£k) as an additional input and Quality Rating (1-10) as an additional output. This is equivalent to 3 inputs and 2 outputs:

Inputs: Person-days, CPU time, Budget (£k)

Outputs: Profit (£m), Quality (1–10). The argument is that a simple model which involves only three measures will overlook certain things. The reality is that, budget constraints are real, and quality of deliverables is not limited to profit only. The addition of more input-output dimensions provides the DMUs with an opening where they can demonstrate efficiency and that is congruent with DEA theory (Banker et al., 1984).

B18 : X ✓ fx =B17/B16										
Project	Profit	Person-days	CPU	Quality (1-10)	Budget (£k)		VIRTUAL	SIGN	VIRTUAL	
P1	2.1	550	200	8	820		1.8929	>=	1.0000	
P2	0.5	400	150	4	710		1.4375	>=	0.5000	
P3	3	300	400	9	520		1.1250	>=	1.1250	
P4	2	350	450	7	660		1.2786	>=	0.8750	
P5	1	450	300	5	560		1.4993	>=	0.6250	
P6	1.5	500	150	7	790		1.7446	>=	0.8750	
P7	0.6	350	200	4	750		1.3268	>=	0.5000	
P8	1.8	200	600	8	880		1.0000	>=	1.0000	
DECISION VARS										
Optimal Weights	W_pday	W_cpu	W_budget	W_profit	W_quality					
	0.00264286		0	0.000535714	0	0.125				
Target DMU										
Select Project:	8									
Target input:	1	=								
Target Output	1	<=								
Efficiency	1.0000									
Project 1	1.0000	Efficient								
Project 2	0.6596	Inefficient								
Project 3	1.0000	Efficient								
Project 4	0.7306	Inefficient								
Project 5	0.6755	Inefficient								
Project 6	1.0000	Efficient								
Project 7	0.6278	Inefficient								
Project 8	1.0000	Efficient								
NOTES — P1 Q2										
1	Cell	Note								
1	K1	Uses stale weights from last Solver run. See summary table below for correct results								
2	B12	Weights from last Solver run (P8). Each project has its own optimal weights – see Answer Reports								
3	B15	Change this & re-run Solver for each project								
4	B18	0 for currently selected project only								
5	A20	Enhanced DEA: 3 inputs (person-days, CPU, budget), 2 outputs (profit, quality). P1,P3,P8,P8 efficient								
6	B20	Correct 9 values from Answer Reports 1-8								
7	B21	0 from Answer Report 1. P1 is efficient (0=1)								
8	B26	0 from Answer Report 6. P6 is efficient (0=1)								

Figure 5: P1 Q4 sheet: improved data with 5 measures

The Solver was also run individually in the 8 DMUs to enhance the performance of the advanced dataset which also utilises the same CCR model.

Project	Efficiency (θ)	Status
P1	1.0000	Efficient
P2	0.6596	Inefficient
P3	1.0000	Efficient
P4	0.7306	Inefficient
P5	0.6755	Inefficient
P6	1.0000	Efficient
P7	0.6278	Inefficient
P8	1.0000	Efficient

The efficient projects were also raised to four. To some degree, this is justified by the fact that the additional dimensions mean one of the DMUs will succeed in at least one of the areas. The quality-to-budget ratio within P6 is also high, whereas low person-days within the P8 in the context of outputs are high. Therefore, a broader portfolio recommendation exists in the improved model, which is P3 profitability and P8 resource efficiency and P6 quality project.

5. Question 5

In order to make decent benchmark comparisons, DEA comparisons across industries need to contain the following:

1. Standardise: Inputs and/Outputs (which include NPV), and or ROI and/or customer satisfaction rating (Ramanathan, 2003).
2. Normalise Data: In order to mitigate differences in scales, z-score normalisation or min-max normalisation can be applied (Bowlin, 1998).
3. Apply BCC Model: The Variable Return to Scale model is better when comparing industries because the companies might be running on various levels (Banker et al., 1984).
4. Carry out Sensitivity Analysis: Determine how robust the rankings are in other specifications of the model and sets of measurements (Saltelli et al., 2008).
5. Validation: Seek professionals in the domain and ask them to confirm that chosen measures/comparisons are not meaningless (Zhu, 2014).

The steps give sound DEA results and enable the ranking of efficiency of industries to be compared with confidence.

Project 2: Linear Programming- Transportation Problem.

1. Question 1

The transportation issue was formulated as a standard linear programming model to minimize Shipping costs from origins to destinations (Taha,2017).

Goal: Minimise $Z = 200x_1 + 800x_2 + 500x_3 + 500y_1 + 400y_2 + 1000y_3$.

Subject to Constraints:

Demand: $x_1 + x_2 + x_3 = 3200$ (steel); $y_1 + y_2 + y_3 = 1000$ (iron).

Capacity: $x_1 + y_1 \leq 2000$; $x_2 + y_2 \leq 1500$; $x_3 + y_3 \leq 2500$.

Non-negativity: all variables ≥ 0 .

Table of costs, plan of shipment, the total cost. Solver Excel (Simplex LP) provides you with the ideal allocation:

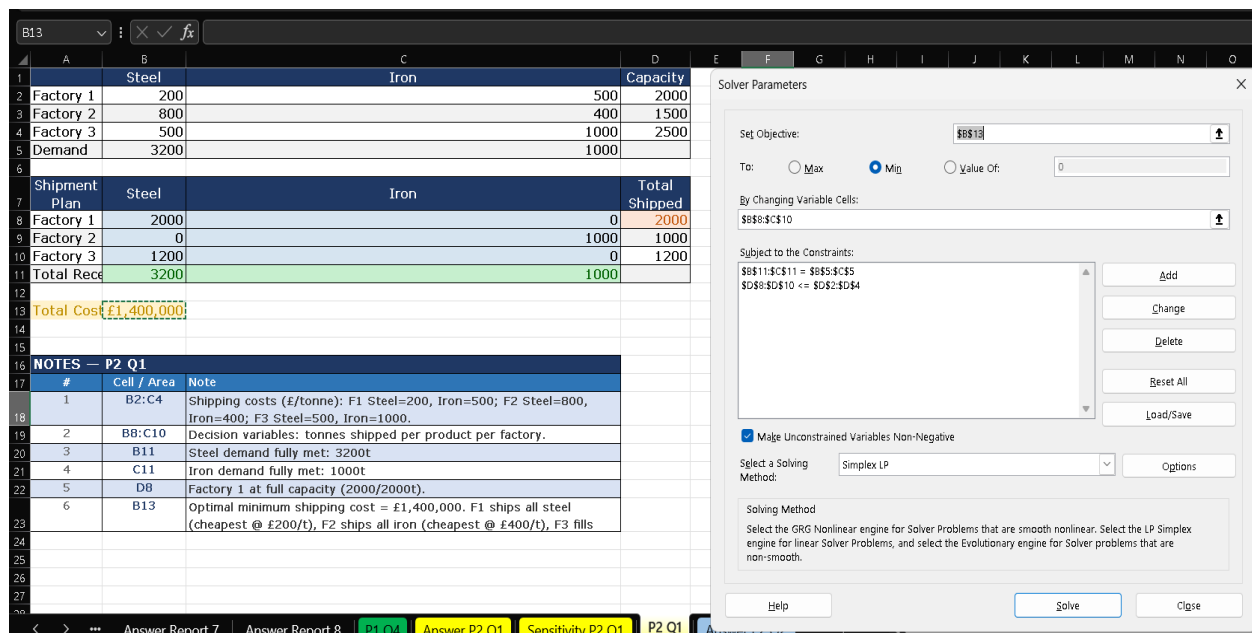


Figure 6: P2 Q1 sheet: cost table, shipment plan, and total cost

Factory	Steel (t)	Iron (t)	Total Shipped(t)
F1	2,000	0	2,000 (at capacity)
F2	0	1,000	1,000

F3	1,200	0	1,200
----	-------	---	-------

Minimum shipping cost: £1,400,000.

Minimum cost of shipping: £1,400,000. The answer is that we can initially put all of the steel in F1 (costing £200/t), iron all in F2 (£400/t), and the remaining is in entirety of steel in 1 200t F3 (£500/t).

A1

Microsoft Excel 16.0 Answer Report

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Worksheet: [Operations Individual Assignment.xlsx]P2 Q1

Report Created: 3/12/2026 3:19:15 PM

Result: Solver found a solution. All Constraints and optimality conditions are satisfied.

Solver Engine

Engine: Simplex LP

Solution Time: 0.016 Seconds.

Iterations: 7 Subproblems: 0

Solver Options

Max Time Unlimited, Iterations Unlimited, Precision 0.000001, Use Automatic Scaling

Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Min)

Cell	Name	Original Value	Final Value
\$B\$13	Total Cost Steel	1700000	1400000

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$B\$8	Factory 1 Steel	1000	2000	Contin
\$C\$8	Factory 1 Iron	0	0	Contin
\$B\$9	Factory 2 Steel	0	0	Contin
\$C\$9	Factory 2 Iron	1000	1000	Contin
\$B\$10	Factory 3 Steel	2200	1200	Contin
\$C\$10	Factory 3 Iron	0	0	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$B\$11	Total Received Steel	3200	\$B\$11=\$B\$5	Binding	0
\$C\$11	Total Received Iron	1000	\$C\$11=\$C\$5	Binding	0
\$D\$8	Factory 1 Total Shipped	2000	\$D\$8<=\$D\$2	Binding	0
\$D\$9	Factory 2 Total Shipped	1000	\$D\$9<=\$D\$3	Not Binding	500
\$D\$10	Factory 3 Total Shipped	1200	\$D\$10<=\$D\$4	Not Binding	1300

Figure 7: Answer Report of P2 Q1.

Microsoft Excel 16.0 Sensitivity Report						
Worksheet: [Operations Individual Assignment.xlsx]P2 Q1						
Report Created: 3/12/2026 3:19:15 PM						
Variable Cells						
Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$8	Factory 1 Steel	2000	0	200	300	1E+30
\$C\$8	Factory 1 Iron	0	400	500	1E+30	400
\$B\$9	Factory 2 Steel	0	300	800	1E+30	300
\$C\$9	Factory 2 Iron	1000	0	400	400	1E+30
\$B\$10	Factory 3 Steel	1200	0	500	300	300
\$C\$10	Factory 3 Iron	0	600	1000	1E+30	600
Constraints						
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$B\$11	Total Received Steel	3200	500	3200	1300	1200
\$C\$11	Total Received Iron	1000	400	1000	500	1000
\$D\$8	Factory 1 Total Shipped	2000	-300	2000	1200	1300
\$D\$9	Factory 2 Total Shipped	1000	0	1500	1E+30	500
\$D\$10	Factory 3 Total Shipped	1200	0	2500	1E+30	1300

Figure 8: Sensitivity Report of P2 Q1.

2. Question 2

The factory 3 is not available (capacity = 0): the overall capacity decreases to 3,500 tonnes (F1: 2,000 + F2: 1,500), and the overall demand is the same (4,200 tonnes required (3,200 steel and 1,000 iron). The deficit of 700 tonnes is deficit of demand and the company is unable to provide the full demand which is needed. Objective Function: Minimise $Z = 200x_1 + 800x_2 + 500y_1 + 400y_2$.

Subject to Constraints:

- Demand: $x_1 + x_2 = 3200$; $y_1 + y_2 = 1000$.
- Capacity: $x_1 + y_1 \leq 2000$; $x_2 + y_2 \leq 1500$; F3 total = 0.
- Non-negativity: all variables ≥ 0 .

When the equality demand is used; the Excel Solver reports as follows: Solver has failed to solve the problem successfully, as the total required demand exceeds available capacity, a fundamental violation of feasibility in transportation models (Taha, 2017). All

the three capacity constraints are binding as confirmed by the Feasibility Report to conclude that the bottleneck has been total capacity.

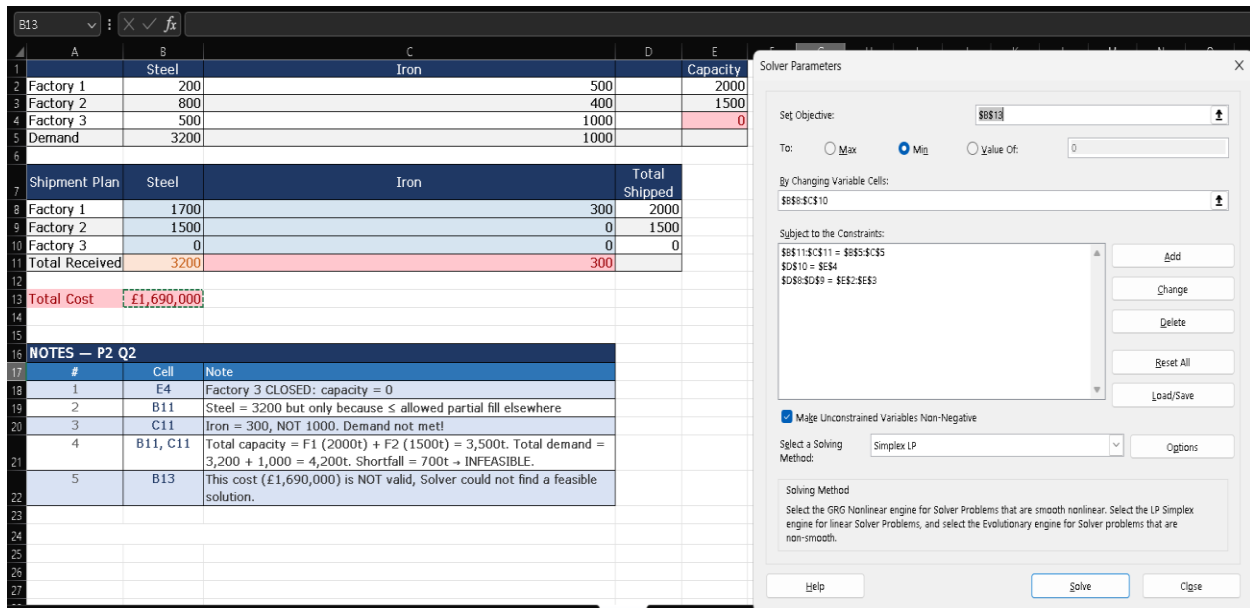


Figure 9: P2 Q2 sheet showing F3 capacity = 0

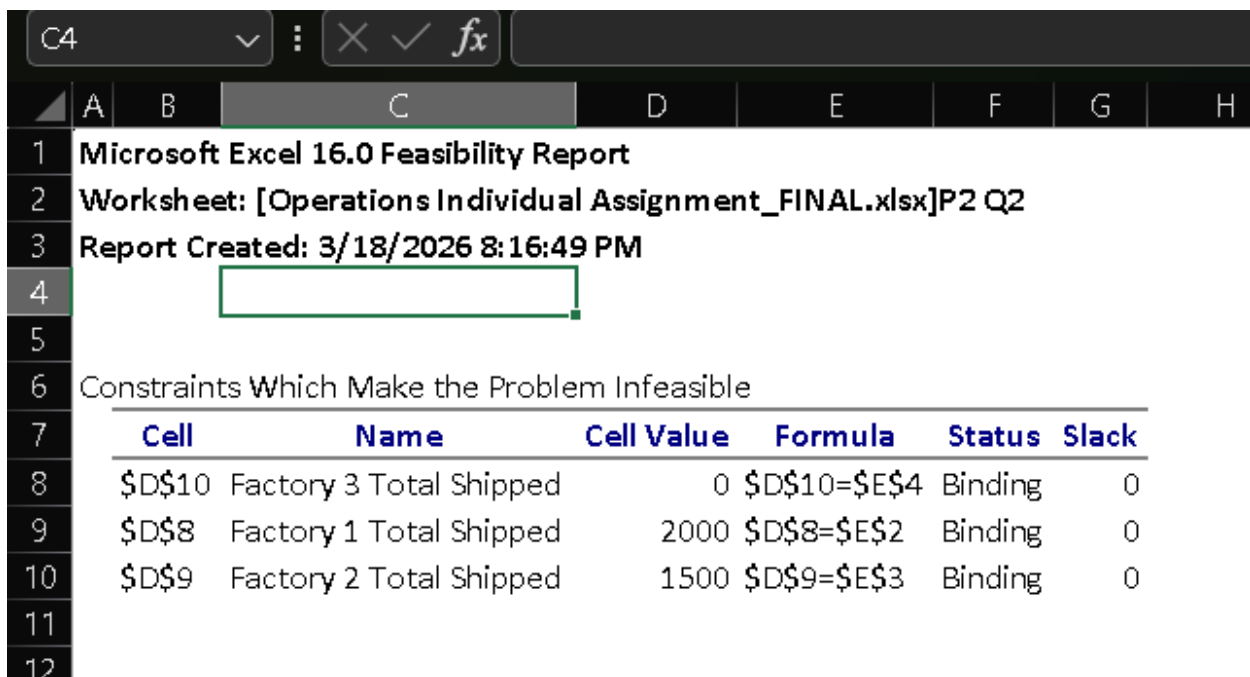


Figure 10: Feasibility Report 2 confirming infeasibility

With a slackened demand ($\leq$ rather than $=$) the optimal partial allocation will be F1: 2,000t steel, F2: 500t steel + 1,000t iron, which will meet all the demand in iron but fail to satisfy

the 700t steel - at £1,200,000.

3. Question 3

The total cost per tonne (shipping + raw materials) is:

Factory	Steel (£/t)	Iron (£/t)
F1	250 (200+50)	600 (500+100)
F2	870 (800+70)	520 (400+120)
F3	545 (500+45)	1,130 (1,000+130)

Refined Objective Function: Minimise $Z = 250x_1 + 870x_2 + 545x_3 + 600y_1 + 520y_2 + 1130y_3$.

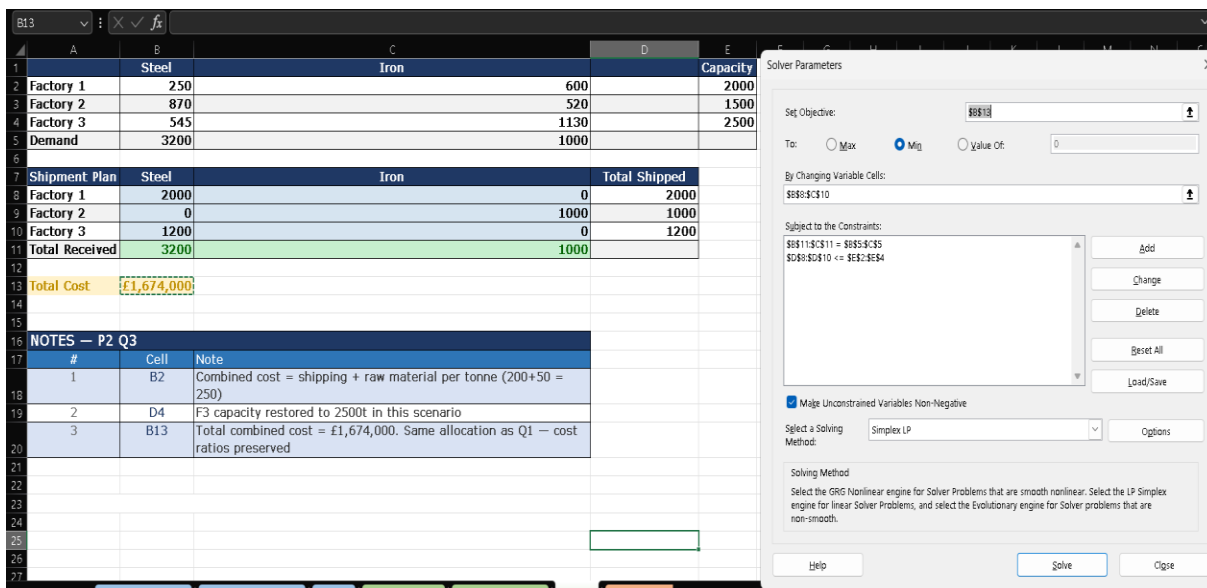


Figure 11: P2 Q3 sheet with combined costs

Optimal allocation will remain the same: But F1 will ship 2,000t steel, F2 will ship 1,000t iron, and F3 will ship 1,200t steel. The aggregate cost is £1,674,000. This is predictable, since the comparative costs of the factories are maintained after taking into account the costs of the raw materials (F1 remains economical with steel (250<545<870), F2 remains economical with iron (520<600<1,130). Simplex solution is going to remain on the same vertex since the ranking is not changing.

A1

Figure 12: Answer Report of P2 Q3

Microsoft Excel 16.0 Sensitivity Report						
Worksheet: [Operations Individual Assignment.xlsx]P2 Q3						
Report Created: 3/12/2026 3:36:17 PM						
Variable Cells						
Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$8	Factory 1 Steel	2000	0	250	295	1E+30
\$C\$8	Factory 1 Iron	0	375	600	1E+30	375
\$B\$9	Factory 2 Steel	0	325	870	1E+30	325
\$C\$9	Factory 2 Iron	1000	0	520	375	1E+30
\$B\$10	Factory 3 Steel	1200	0	545	325	295
\$C\$10	Factory 3 Iron	0	610	1130	1E+30	610
Constraints						
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$B\$11	Total Received Steel	3200	545	3200	1300	1200
\$C\$11	Total Received Iron	1000	520	1000	500	1000
\$D\$8	Factory 1 Total Shipped	2000	-295	2000	1200	1300
\$D\$9	Factory 2 Total Shipped	1000	0	1500	1E+30	500
\$D\$10	Factory 3 Total Shipped	1200	0	2500	1E+30	1300

Figure 13: Sensitivity Report for P2 Q3

Project 3: Network Optimisation

1. Question 1

The problem of shortest path has been represented as a network flow LP, there were 12 network nodes (0-11), and 16 directed arcs. Binary variables show whether all the arcs are on the path or not, and the optimal path between two nodes is obtained. (+1 at the source, - 1 at sink, 0 otherwise).

Best path: 0 -1 -5- 7- 11 Minimum cost: £1,150 (300 + 460 + 320 + 70)

In the context of digital marketing, it implies that the brand must invest in the chain of influence using these four influencers and, in the process, it will minimise the overall cost by lowering the cost of going viral in the source (node0) to the target (node11).

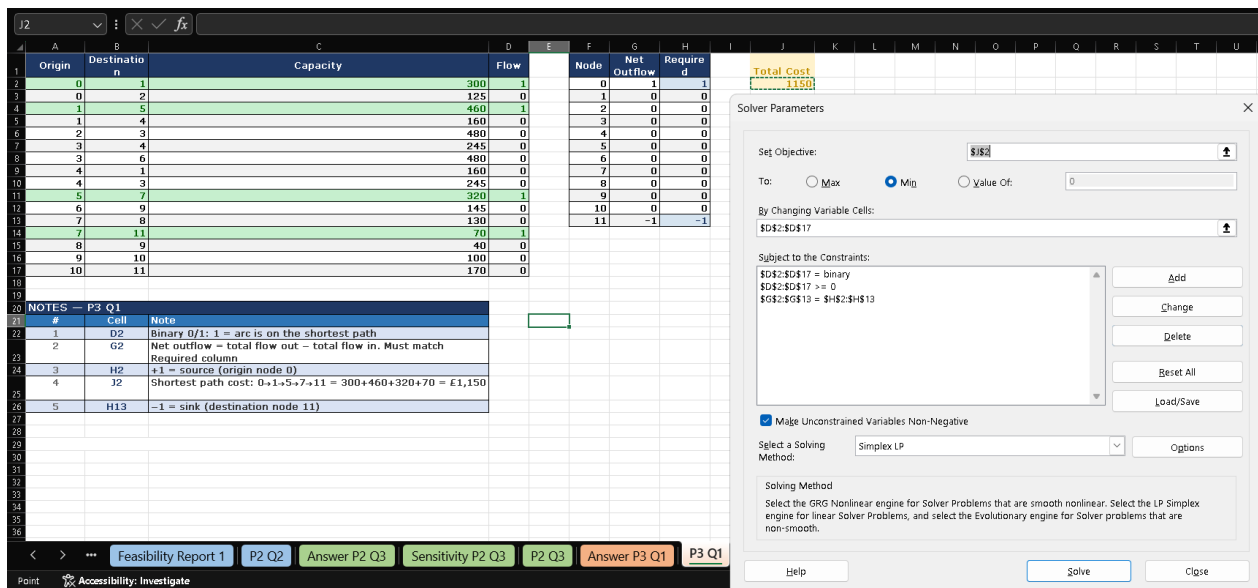


Figure 14: P3 Q1 Sheet: Arc Table with Binary Flow

A1

Figure 15: P3 Q1 sheet arc table with binary decision variables

2. Question 2

a. Optimal Rollout (9 marks):

Between Node 0 to Node 11, the total data throughput needs to be maximised to formulate the maximum flow problem. Arc capacities limit the continuous flow variables, the flows in the intermediate nodes are maintained constant.

Top possible flow: 170 units through 2 parallel routes:

From_Node	To_node	Capacity	Flow (Decision Variables)	Node	Net flow	Required	Total Outflow
0	1	300	70	1	0	0	170
0	2	125	100	2	0	0	
1	5	460	70	3	0	0	
1	4	160	0	4	0	0	
2	3	480	100	5	0	0	
3	4	245	0	6	0	0	
3	6	480	100	7	0	0	
4	1	160	0	8	0	0	
4	3	245	0	9	0	0	
5	7	320	70	10	0	0	
6	9	145	100				
7	8	130	0				
7	11	70	70				
8	9	40	0				
9	10	100	100				
10	11	170	100				

Path 1 (70 units): Node 0 sends 70 units to Node 1 → Node 5 → Node 7 → Node 11.
 Path 2 (100 units): Node 0 sends 100 units to Node 2 → Node 3 → Node 6 → Node 9 → Node 10 → Node 11.

The pipe from Node 7 to Node 11: The flow is 70, and its maximum capacity is 70.
 The pipe from Node 9 to Node 10: The flow is 100, and its maximum capacity is 100.

#	Cell	Note
1	D2	Flow on each arc (decision variable). Must be ≤ Capacity
2	G2	Flow conservation: inflow = outflow at each intermediate node
3	J2	Max flow from node 0 to node 11 = 170 units
4	F13	Path 1: 0→1→5→7→11 carries 70 units
5	D14	Bottleneck: 7→11 at full capacity (70/70)
6	F14	Path 2: 0→2→3→6→9→10→11 carries 100 units
7	D16	Bottleneck: 9→10 at full capacity (100/100)

Figure 16: P3 Q2 sheet: flow values and max flow result

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method: Simplex LP

Solving Method
 Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Buttons: Add, Change, Delete, Reset All, Load/Save, Help, Solve, Close

Figure 17: Solver setup for P3 Q2

Maximum flow: 170 units, via two parallel paths:

- Path 1: 0 – 1 – 5 – 7 – 11 – carrying 70 units.
- Path 2: 0 – 2 – 3 – 6 – 9 – 10 – 11 – carrying 100 units.

The total of all the outflows of Node 0 is a figure of 170 units. This provides the throughput of all the capacity available between Node 0 and Node 11 in a single deployment.

b. Bottlenecks (3 marks):

The Max-Flow Min-Cut Theorem is that maximum flow is equal to minimum cut capacity. Two arcs are filled in (capacity = flow, slack = 0):

Arc 7 - 11: capacity 70, flow 70 constraints Path 1.

Arc 9 - 10: capacity 100, flow 100 hampers Path 2.

This is the network bottleneck. A non-zero shadow price on this constraint has been indicated in the sensitivity report, so it would mean that by increasing the capacity of arc 7-11 the maximum flow would be increased. To the game company, the bandwidth constraints are what they should give priority and upgrade on.

Report Created: 3/13/2026 10:14:29 PM							
Variable Cells							
Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease	
\$D\$2	Flow (Decision Variables	70	0	1	0	0	
\$D\$3	Flow (Decision Variables	100	0	1	0	0	
\$D\$4	Flow (Decision Variables	70	0	0	0	1	
\$D\$5	Flow (Decision Variables	0	0	0	0	0	
\$D\$6	Flow (Decision Variables	100	0	0	0	0	
\$D\$7	Flow (Decision Variables	0	0	0	0	1E+30	
\$D\$8	Flow (Decision Variables	100	0	0	1E+30	0	
\$D\$9	Flow (Decision Variables	0	0	0	0	1E+30	
\$D\$10	Flow (Decision Variables	0	0	0	0	1E+30	
\$D\$11	Flow (Decision Variables	70	0	0	0	1	
\$D\$12	Flow (Decision Variables	100	0	0	1E+30	0	
\$D\$13	Flow (Decision Variables	0	0	0	0	1E+30	
\$D\$14	Flow (Decision Variables	70	1	0	1E+30	1	
\$D\$15	Flow (Decision Variables	0	0	0	0	1E+30	
\$D\$16	Flow (Decision Variables	100	1	0	1E+30	1	
\$D\$17	Flow (Decision Variables	100	0	0	1E+30	1	
Constraints							
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease	
\$G\$2	Net flow	0	1	0	230	70	
\$G\$3	Net flow	0	1	0	25	100	
\$G\$4	Net flow	0	1	0	25	100	
\$G\$5	Net flow	0	1	0	160	0	
\$G\$6	Net flow	0	1	0	230	70	
\$G\$7	Net flow	0	1	0	25	100	
\$G\$8	Net flow	0	1	0	230	70	
\$G\$9	Net flow	0	1	0	130	0	
\$G\$10	Net flow	0	1	0	25	100	
\$G\$11	Net flow	0	0	0	100	70	

Figure 18 - Sensitivity Report marked with binding constraints.

3. Question 3

Shortest Path Applications:

1. Logistics and Supply Chain: Route Optimisation

Delivery companies such as Amazon and FedEx are using the idea of shortest-path algorithms (Dijkstras algorithm) to select the most suitable route that uses the least amount of fuel and shortest possible transit time. The node is a warehouse or depot and the arc is a road and the distance or driving time is the weight of the edge (Ahuja et al., 1993). This conserves the fuel costs, wear and tear on vehicles and improves delivery times.

2. Telecommunications: Use of Data Packet Routing

Internet service providers employ a shortest-path algorithm (e.g. OSPF which implements the algorithm of Dijkstra) to transmit data packets through network routers. This is achieved by reducing latency, reduced lost data and load balancing of the network. In the case of cloud-based service provider, effective routing is an evident competitive edge.

Maximum Flow Applications:

1. Manufacturing: Assembly Line Balancing

An assembly line may be modelled as a network with each work station modelled as a node with the capacity of the arcs set to the maximum number of units that can be processed in a specific hour. The maximum flow model defines a hypothetical maximum throughput and the workstations which act as bottlenecks (the minimum cut). The management can then focus on investments to fill such bottlenecks as a way of increasing the total factory production and the final revenue (Ford and Fulkerson, 1956).

2. Finance and Project Management: Capital Budgeting

A complicated project may be modelled as a network with the milestones as nodes and the dependencies as arcs. Capacity of an arc refers to the level of the capital or human resources that are available each quarter. This maximum flow model is an effective resource allocation tool by ensuring that the maximum rate a project can be completed and the phase that is stalling progress.

References

- Ahuja, R. K., Magnanti, T. L., & Orlin, J. B. (1993). *Network flows: Theory, algorithms, and applications*. Pearson.
- Banker, R. D., Charnes, A., & Cooper, W. W. (1984). Some models for estimating technical and scale inefficiencies in data envelopment analysis. *Management Science*, 30(9), 1078–1092. <https://doi.org/10.1287/mnsc.30.9.1078>
- Bowlin, W. F. (1998). Measuring performance: An introduction to data envelopment analysis. *Journal of Cost Analysis*, 15(2), 3–27.
- Charnes, A., Cooper, W. W., & Rhodes, E. (1978). Measuring the efficiency of decision making units. *European Journal of Operational Research*, 2(6), 429–444. [https://doi.org/10.1016/0377-2217\(78\)90138-8](https://doi.org/10.1016/0377-2217(78)90138-8)
- Cooper, W. W., Seiford, L. M., & Tone, K. (2007). *Data envelopment analysis: A comprehensive text* (2nd ed.). Springer.
- Ford, L. R., & Fulkerson, D. R. (1956). Maximal flow through a network. *Canadian Journal of Mathematics*, 8, 399–404. <https://doi.org/10.4153/CJM-1956-045-5>
- Hillier, F. S., & Lieberman, G. J. (2021). *Introduction to operations research* (11th ed.). McGraw Hill.
- Ramanathan, R. (2003). *An introduction to data envelopment analysis*. Sage Publications.
- Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., & Tarantola, S. (2008). *Global sensitivity analysis: The primer*. John Wiley & Sons.
- Taha, H. A. (2017). *Operations research: An introduction* (10th ed.). Pearson.
- Zhu, J. (2014). *Quantitative models for performance evaluation and benchmarking* (3rd ed.). Springer.